A FORM OF MYASTHENIA GRAVIS WITH CHANGES IN THE CENTRAL NERVOUS SYSTEM.

BY DOUGLAS McALPINE, M.D.

(From the Pathological Laboratory of the Hospital for Epilepsy and Paralysis, Maida Vale, London.)

THIS paper is based on a clinical and pathological account of a case of myasthenia gravis. Certain morbid changes were found in the central nervous system which have not previously been described.

Clinical Account of the Case.

G. E. H., a probationer nurse, aged 22, was admitted to the Hospital for Epilepsy and Paralysis, Maida Vale, London, on February 15, 1924, under the care of Dr. H. Campbell Thomson, who subsequently handed the case over into my care with a diagnosis of myasthenia gravis.

History.—The patient had enjoyed good health until October, 1923, when she noticed a slight difficulty in articulating certain words. A month later the lower limbs became weak after standing for any length of time. In December the arms were similarly affected. During the first week in January, 1924, she saw double for the first time. The patient particularly noticed that all her symptoms were more marked towards the evening, or after much exertion. In January she developed a right otitis media which was followed by an intermittent discharge from the ear. No history of insomnia, somnolence, pain or involuntary movements could be elicited. As far as the patient was aware she had had no rise of temperature, except possibly when the ear first discharged. There was nothing of importance in the family or personal histories.

Condition on February 18, 1924.—The patient appeared to be a healthy young woman, though inclined to stoutness. She was intelligent and able to give a clear account of her illness. The pupils were equal and reacted well to light and accommodation. There was a slight but definite impairment of ocular movements in a lateral direction. Diplopia could be easily induced by asking the patient to perform repeated lateral movements of the eyes. Upward movement of the eyeballs was slightly restricted. Moderate bilateral ptosis was present, which varied in degree from day to day, but which was always more marked towards the evening. The facial muscles were bilaterally weak, more especially the orbicularis oculi. The circumoral muscles were less affected but the degree of paresis was sufficient to prevent the act of whistling, which had been possible before her illness. The masseters and temporal muscles contracted well and were not atrophied. The voice was considerably altered; it was feeble in tone and somewhat monotonous; at times dysarthria was marked, rendering her speech difficult of comprehension. The palate moved well on phonation but was easily fatigued. The tongue was protruded well and showed no atrophy or tremor; the patient had difficulty, however, in keeping it protruded for any length of time. The sterno-mastoid muscles and upper portion of the trapezii were not atrophied or weak, but both they and the extensor and flexor muscles of the neck tired easily.

Upper limbs.—Muscular atrophy was absent. Motor power was quite good, but repeated movements, especially at the shoulder, were quickly accompanied by lessened force and a subjective sensation of fatigue. The reflexes were brisk and equal on the two sides. There was no sensory loss.

Trunk.—No paresis of the abdominal or intercostal musculature could be detected. The abdominal reflexes were brisk and equal on the two sides.

Lower limbs.—Motor power was relatively good. The knee-jerks and ankle-jerks were brisk. In the case of the knee-jerk, taps on the patellar tendon repeated at short intervals resulted in a definite decrease in the degree of extension. Both plantar responses were flexor in type. There was no sensory loss. Gait was normal.

Electrical reactions.—A fairly typical myasthenic reaction was obtained in the fore-arm muscles.

Sphincters.—Control over these was normal.

Heart.—This was not enlarged and the heart sounds were normal.

Abdomen.—The liver and spleen were not enlarged.

Urine.—This was free from albumin and sugar.

Endocrine glands.—The thyroid gland did not appear enlarged. No definite enlargement of the thymus gland could be made out either by percussion or on X-ray examination. Menstruation was normal. There was no pigmentation of the skin and the blood-pressure figures were: Systolic 125, diastolic 75.

Ear, nose and throat.—There was a right otitis media, with a purulent discharge from this ear. The tonsils were enlarged and appeared septic.

Serological findings.—The blood Wassermann reaction was negative. The cerebro-spinal fluid was clear and was not under tension. The Wassermann reaction was negative. There were 3-5 cells per c.mm. and 0·035 per cent. of protein. The Lange gold curve was normal.

Progress.—Treatment of the otitis media was unsatisfactory, and on this account it was decided to remove the tonsils. These were dissected out on July 19. During many months in hospital her condition slowly grew worse. The muscular weakness in her limbs and elsewhere, which at first was only apparent after exertion, became permanent. The neck muscles, towards the end of her life, were considerably affected, so that at times, especially towards the evening, she was forced to support her chin with her hands. Dysarthria became permanent and latterly her speech was nearly incomprehensible. Dysphagia appeared about four months before death, and to this was added difficulty in mastication. Diplopia with partial external ophthalmoplegia were constantly present. The most distressing symptom, however, was dyspnœa, which was first noted in November, 1924. This occurred in attacks and was sometimes brought on by crying, which seemingly fatigued the diaphragm and intercostal muscles. The attacks lasted sometimes as long as four hours. The reflexes remained unaltered throughout the illness. On the morning of January 21, 1925, she had a severe attack of dyspnœa which persisted throughout the day with full retention of consciousness. Respiration was exceedingly shallow and difficult. Towards the evening she became comatose and died a few hours later.

Clinical summary.—A female, aged 23, developed dysarthria in October, 1923, followed a month later by diplopia. She readily became fatigued. These symptoms were always more marked towards the evening or after exertion. There was no history of sleep disturbance, pains or involuntary movements. When examined in February, 1924, she showed the following signs: A moderate bilateral and variable ptosis, diplopia and marked weakness of the orbicularis oculi and to a less extent of the lower facial musculature. Fatigue was easily induced in the soft palate, tongue and limbs. The reflexes and sensation were normal. During many months in hospital her condition gradually grew worse; her symptoms were always worse towards the end of the day. Dysphagia, weakness of the neck muscles and permanent paresis of the limbs with attacks of dyspnœa became marked, and she died in a respiratory attack in January, 1925, her illness having lasted fifteen months.

Pathological Findings.

At the post-mortem examination, performed seven hours after death, no abnormality of the brain or spinal cord was noted. The thymus gland was much enlarged and weighed in the fresh state 48·5 grm. Parts of the following organs were removed for histological examination: thymus and thyroid glands, liver, spleen, kidney, suprarenal gland, uterus, Fallopian tube, ovary and heart. Portions of certain muscles were also removed, e.g., tongue, clavicular head of left sterno-mastoid, right external rectus, diaphragm, upper part of right trapezius, right biceps and deltoid.

Histological examination.—The various internal organs appeared normal with the exception of the thymus. The general structure of the gland was normal, but there was considerable increase in the depth of the cortex. The medulla contained about a normal proportion of Hassall's corpuscles. Owing to the kindness of Professor J. McIntosh, I was able to compare the sections with those from a young woman, aged 20, who suddenly died whilst dancing and in whom the only post-mortem finding of importance was an enlarged thymus gland. The histological appearances in the two cases were practically identical and pointed to a simple hypertrophy.

Numerous sections of the muscles mentioned above were carefully examined with completely negative results. No collection of cells resembling lymphorrhages were seen.

Nervous system.—The brain and spinal cord were fixed in a 10 per cent. solution of formalin. Various parts of the cortex, the basal ganglia and cerebellum were examined.

The brain-stem from the subthalamic region downwards was cut in serial section. Many sections of the spinal cord at various levels were studied. The methods used were: hæmatoxylin and van Gieson, hæmatoxylin and eosin, Nissl, Weigert-Pal, Loyez, Anderson's neuroglial stain, Scharlach R., Bielschowsky and Mayer's mucicarmine.

Cortex.—Sections from the frontal, rolandic, parietal and occipital regions were examined. The cells of the various layers appeared normal. The vessels were normal except for deposits of mucin in the perivascular spaces. The Weigert-Pal preparations showed normal staining of the fibres, except when they were interrupted by mucocytes, which had pushed the fibres aside, leaving clear rounded spaces of varying size. The meninges were normal except for infiltration with mucoid material; this will be referred to later.

Basal ganglia.—The cells of the various nuclei appeared plentiful and the great majority were normal. There was no evidence of perivascular lymphocytosis. The fibres of the internal capsule stained well but were interrupted at frequent intervals by mucocytes of varying size.

Subthalamic region.—The various nuclei, such as the corpus Luysii, red nucleus, substantia nigra, tuber cinereum, and the nuclei in the neighbourhood of the third ventricle were normal, and no changes were seen in the vessels or neuroglial cells.

Brain-stem.—The nuclei of the cranial nerves were normal with the exception of the third nerve nucleus, a few cells of which showed some variation in their staining qualities but no definite evidence of chromatolysis. No intracellular lipoid material was seen. The vessels and meninges throughout the brain-stem were free from any cellular infiltration, except at the junction of the medulla and pons, where a small vessel showed a discrete perivascular lymphocytosis. Mucoid degeneration was present throughout the white matter of the brain-stem.

Spinal cord.—As the changes about to be described were similar in all parts of the cord, the following description of those present in the cervical region is applicable to other parts of the cord. The anterior horn cells were normal. The vessels showed well-marked perivascular infiltration with round cells. This was a striking feature in the majority of the sections examined, four to six rings of cells being commonly seen. The cells were sometimes arranged in the form of a cuff, similar to that seen in epidemic encephalitis. The majority of the cells were lymphocytes of medium size; their nuclei, containing a number of chromatin dots, stained well with hæmatoxylin. The nuclei varied in shape, some being rounded, others oval. Other cells present were small lymphocytes with a more deeply-staining nucleus, and hyaline endothelial cells with a large oval nucleus. Plasma cells were scarce. The perivascular infiltration was most marked near the postero-median fissure and in the region of the posterior root entry zone. In one or two sections cuffing was seen around a vessel in the anterior horn.

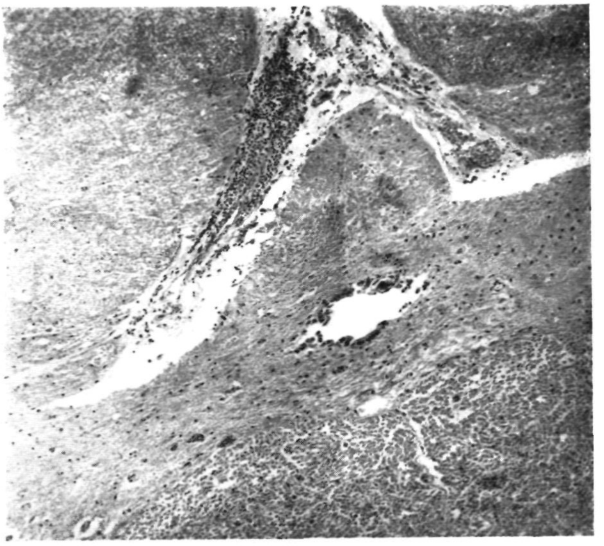

FIG. 1.—Section through mid-dorsal region of the cord, showing perivascular infiltration around a vessel in the antero-median fissure. Hæmatoxylin and eosin. x 60.

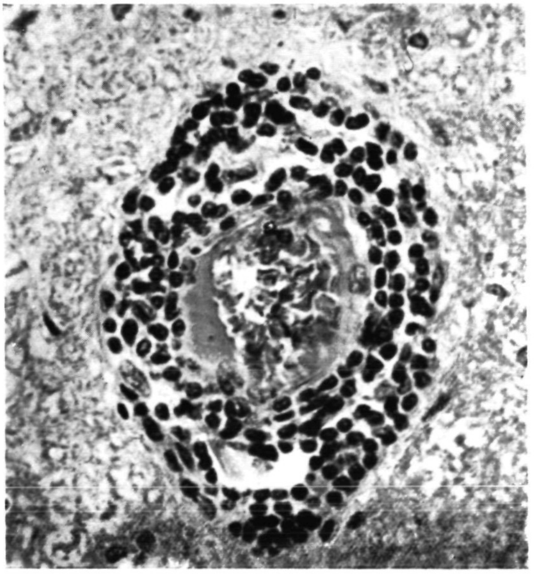

FIG. 2.—A section from the cervical enlargement, showing details of the perivascular infiltration. Hæmatoxylin and eosin. x 480.

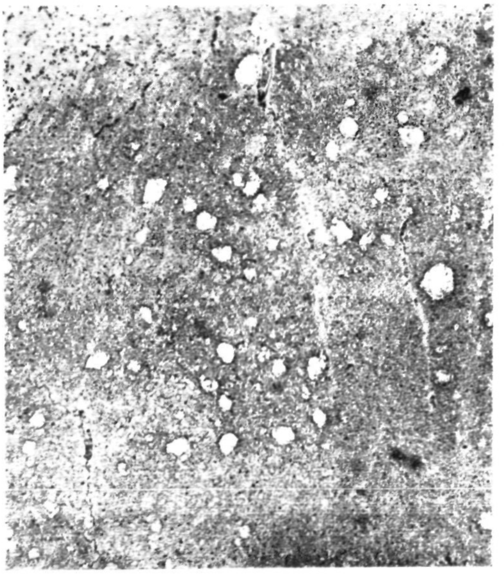

FIG. 3.—Numerous mucocytes in the white matter of the spinal cord. Hæmatoxylin and van Gieson. x 50.

To illustrate Dr. Douglas McAlpine's paper.

Recent hæmorrhages into the grey matter of the anterior horns were observed in several sections, being absent from the brain-stem and brain itself. There was a slight but definite increase of neuroglial cells, chiefly around the anterior horns and posterior root entry zone. The meninges were normal except for deposits of mucoid material beneath them. In celloidin sections small and large mucocytes were numerous throughout the length of the cord.

Weigert-Pal sections showed no tract degenerations and were normal, except for a riddled appearance due to the mucocytes.

Mucoid Degeneration.

As deposits of mucin were extremely numerous, their character and disposition will be considered in more detail. In sections stained by the method of Nissl, especially when thionin blue was used, rounded or irregularly-shaped areas staining a faint pink colour were seen in the white matter. These areas were stained a bright blue colour with hæmatoxylin. The appearances in Weigert-Pal sections have already been described. The nature of this degeneration was made clear by staining sections with Mayer's mucicarmine, which gave a bright pink colour to the areas. This stain has been described as specific for mucin.

In a section through the cortex and adjacent subcortical white matter, the process was observed to be intense in the white matter. The vessels of the cortex itself were in many places surrounded by mucin. A diffuse infiltration of the meninges was observed in many places, and the process tended to invade the adjacent cortex. The general appearance of the degeneration in this region suggested that mucin was transported to the surface of the brain from the subcortical white matter through the agency of the perivascular lymphatics. The nuclei of the basal ganglia were relatively free, but vessels here and there contained mucin in their perivascular spaces. The internal capsule showed a great quantity of mucin scattered about in areas of varying size. Throughout the brain-stem the process was generalized, affecting chiefly the white matter. The degeneration in most places was at an early stage, the areas being comparatively small in size. The process extended up to the meninges and under the ependyma. The cortex of the cerebellum was unaffected, but in its white matter and the dentate nucleus the process was more marked than in the brain-stem or basal ganglia. In the spinal cord the degeneration was particularly intense. It was practically confined to the white matter.

The areas varied considerably in size, some being quite large, due to fusion of several adjacent mucocytes. The mucoid material also tended to collect along the entering posterior root fibres.

Summary of Histological Findings.

The various visceral and endocrine organs were normal, with the exception of the thymus gland which was markedly hypertrophied. No lymphorrhages were observed in the muscles examined. The cerebral cortex, subcortical white matter and basal ganglia were normal apart from extensive mucocytic degeneration, which especially involved the white matter. The various motor nuclei in the brain-stem were unaltered except that a few of the cells of each third nucleus showed slight changes. The vessels were normal. Mucocytic degeneration was present. Many of the vessels of the spinal cord were markedly infiltrated with round cells. There was a slight degree of gliosis, especially around the anterior horns. The motor cells were normal at all levels. Mucocytes were more numerous in the cord than elsewhere. The meninges were normal.

THE SIGNIFICANCE OF MUCOCYTIC DEGENERATION.

Reference has been made to areas of mucoid degeneration. A review of the literature shows that this morbid change has attracted but little attention until quite recently. Buscaino [6], in studying brains from some cases of dementia præcox, found "grape-like areas" of disintegration in the white matter. He considered that these were the result of a true process of disintegration due to a lesion of the nerve fibres and neuroglial cells. Subsequently other workers on this subject found these "grape-like bodies" in a variety of pathological and even normal brains. It was shown that this form of degeneration was not apparent in frozen sections but only in those previously treated with alcohol or nitric alcohol. On this account Salustri [21], Ansalone [1], S. d'Antona [2] and others considered that they were artefacts. Lhermitte and Radovici [14], in cases of epidemic encephalitis, observed rounded areas with clear-cut edges, which stained red by the method of Casamajor, and violet-red by the method of Bonfiglio. The process was most marked in the white matter of the brain and spinal cord. It was sometimes observed in neuroglial cells, but more often it was extracellular and closely connected to the myelin sheaths. Grynfeltt [11], using Mayer's mucicarmine, was the first to recognize the nature of the degeneration and its origin in the interfascicular glial cells. He introduced the word "mucocyte" to indicate the cellular origin and chemical nature of this form of degeneration. His work was based on the examination of a brain of an old man, but he gave no details as to the nature of the disease from which the man suffered. Lhermitte, Kraus and Bertillon [15] described fully the staining reactions of these "mucin-like bodies," which they observed in a case of post-encephalitic Parkinsonism, but denied their cellular origin.

Greenfield [9] observed the formation of mucin or mucinoid droplets in the bodies of interfascicular glial cells in epidemic encephalitis. He considered that the change may take place early, as he had observed it on the twenty-fourth day of the disease. Pélissier [20] studied this process in great detail in a case of chronic encephalitis. He described the distribution of mucocytes in the white matter throughout the brain, and stated that their presence in the grey matter can be explained by a migration of the mucocytes from the white matter. He also found mucoid material under the ependyma. He described in detail the formation of mucocytes from interfascicular glial cells.

Bailey and Schaltenbrand [3] observed this type of degeneration in a case of Schilder's disease. They adopted the term "mucocyte" as used by Grynfeltt and Pélissier, and proved that the interfascicular cells referred to by these authors are oligodendroglial cells. Grynfeltt and Pélissier showed that the mucin, formed for the most part in the white matter, found its way towards the surface of the brain on the one hand, and towards the ventricular cavities on the other. This migration appeared to be carried out through the agency of the perivascular lymphatic spaces.

Pagès, Bénoit and Pèlissier [18] have deduced from these findings the possibility of mucin appearing in the cerebro-spinal fluid in certain diseases. The method which they used for the detection of this substance in the fluid is that of Derrien, the details of which are described in their paper. They have searched for mucin in 130 cases. In fifteen cases in which the fluid was withdrawn during spinal anæsthesia for various operations, no mucin was found. It was present, however, in a variety of organic neurological conditions, such as cerebral arteriosclerosis, epilepsy, dementia præcox and uræmia. It occurred most frequently in cases of chronic epidemic encephalitis.

Ferraro [8], in a recent critical review of the subject, based on his own work and that of others, came to the following conclusions:—

(1) The "grape-like body" of Buscaino and the mucocyte of Grynfeltt and Pélissier are identical.

(2) Their absence from frozen sections does not necessarily imply an

artefact, since the action of alcohol may reveal a form of lesion otherwise not apparent.

(3) The relationship between acute swelling of the oligodendroglia and the formation of mucocytes has not, in his opinion, been satisfactorily demonstrated. The absence of striking changes in the nuclei in the area of degeneration is a point against the origin of mucocytes from degenerated neuroglial cells.

I have attempted to repeat certain of Ferraro's experiments, but without any convincing results. The following are my conclusions, tentatively given : (1) Small rounded areas of mucin are present in frozen as well as celloidin sections of a large number of pathological conditions. No controls with normal material have been made. (2) In formalin-fixed material, which has been passed through celloidin, numerous irregularly-shaped areas corresponding to the large mucocytes of Grynfeltt and Pélissier have been found in the case of myasthenia gravis described in the present paper and in two cases of chronic epidemic encephalitis. These large mucocytes were absent from frozen sections both before and after treatment with alcohol for twenty-four hours. No attempt was made to determine the mode of formation or the origin of the small mucocytes, or the relationship between these and the large mucocytes of Grynfeltt and Pélissier. It seems obvious, however, that whatever their origin the presence of small mucocytes, owing to their frequency in morbid conditions of the brain and cord, is of little importance. On the other hand, the large mucocytes are in my experience much rarer, and may ultimately prove to be of considerable pathological interest.

Myasthenia gravis is a disease whose ætiology is at present unknown. No attempt will be made here to discuss the various pathological and biochemical theories that have been put forward from time to time to explain the curious phenomena so characteristic of the disease. An excellent summary of our present knowledge of the subject will be found in a recent paper by Keschner and Strauss [16]. The most constant pathological changes are collections of small round cells, resembling lymphocytes, in muscles, and certain visceral organs, to which Buzzard [7] gave the name of "lymphorrhages," and an enlargement of the thymus gland, which is hypertrophied or in some cases the seat of a tumour, usually benign. These two findings exist in about half the reported cases, although not always in association. According to the published accounts of the disease, changes in the central nervous system are rare. A number of different observers have noted varying degrees of chromatolysis in the anterior horn cells of the cord. Buzzard, in one of his cases, found a few small collections of lymphocytes in the posterior root ganglia. Mandelbaum and Celler [17] described a lymphocytic infiltration around capillaries in the grey matter near the tenth nerve nucleus and ventral to the olivary body in one case. Barrada [4] laid emphasis on the presence of lipoid granules in the cells, especially of the thalamus and third nerve nucleus. Similar collections of lipoid material, however, are not uncommon in other pathological conditions. The above findings in the nervous system show no uniformity and are so relatively scarce that they can, I think, be safely disregarded in considering the ætiology of the disease.

Myasthenia Gravis as a Possible Sequel to Epidemic Encephalitis.

Such marked inflammatory and other changes in the spinal cord and brain as were present in this case suggest the possibility of an occasional relationship between epidemic encephalitis and myasthenia gravis. The association between these two conditions has been the subject of a few recent papers, and the case reports will now be considered in detail.

Grossman [10], in a paper entitled "Epidemic Encephalitis simulating Myasthenia Gravis," seems to have been the first person to draw attention to the possibility of this association. He described three cases in which the main symptoms and signs were a sudden appearance of diplopia with ptosis and varying degrees of ophthalmoplegia, dysarthria and dysphagia, an easily induced fatiguability of the limbs, and later the development of paresis and wasting, especially of the proximal musculature of the upper limbs. Two cases showed early bladder involvement and a doubtful plantar response, whilst one case exhibited choreiform movements. Grossman comments on the apparently acute onset of the illness, the lack of variation in the degree of muscular weakness, the absence of a myasthenic electrical reaction and the persistent ophthalmoplegia, as points against the diagnosis of a primary myasthenia gravis. On the other hand, the vesical disturbance and the doubtful plantar response in two of the cases suggested epidemic encephalitis. In one case that came to autopsy there was marked congestion of the vessels in the medulla with lesions of subacute inflammatory nature such as perivascular lymphocytosis and multiple hæmorrhages, most intense in the mid-brain.

Sarbo [22] cites the case of a male, aged 32, who had an attack of epidemic encephalitis in 1920. Bilateral facial paralysis with lethargy lasting five weeks were the principal signs. Subsequently, he developed a rapid fatiguability of all his musculature. This was obvious during every movement, but disappeared if he lay down. The myasthenic electrical reaction was obtained on more than one occasion. No mention is made of the presence or absence of Parkinsonism. Paulian [19] describes a case occurring in a female, aged 29. In December, 1921, she was taken ill with fever and somnolence which lasted two months. This was succeeded by muscular weakness and intermittent diplopia, difficulty in walking, chewing and swallowing. Food regurgitated through her nose. The myasthenic electrical reaction was obtained. She presented the classical picture of myasthenia gravis, and this diagnosis was confirmed by several noted neurologists. She improved for a time, but, whilst abroad, she contracted pneumonia and died. The authors consider that the diplopia and somnolence at the onset pointed to an attack of epidemic encephalitis.

Wimmer and Vedmand [23] report two cases. The first was that of a male, aged 42, who was taken ill in October, 1918, with fever and somnolence. There were no ocular or other symptoms. The condition lasted for about three weeks. Within the next few months he noticed muscular fatiguability, nasal speech and dysphagia, followed by a falling of the lower jaw, which he was forced to support with his hand. In September, 1919, the speech became nearly unintelligible, after reading in a loud voice for one minute. The myasthenic electrical reaction was positive. In 1920, weakness in the arms was noted. He continued at work up till August, 1925, when he was compelled to enter hospital again. He showed atony and weakness of the facial musculature, and especially of the orbicularis oculi, with diplopia due to weakness of the left external rectus. There was atrophy of the masseters and tongue; the latter could not be protruded. Speech was indistinct and at times incomprehensible. There was a slight generalized wasting of the musculature of the limbs with a reduction in power. The faradic and galvanic electrical responses were lost in the masseters and temporal muscles, but no myasthenic reaction could now be obtained. During 1926 the patient improved.

The second case was that of a male, aged 40, who was taken ill with diplopia, muscular fatigue and bulbar symptoms. Somnolence and fever were absent. He improved, and continued at his work until August, 1924, when he noticed that his voice became weak after talking for a time, except in the morning. On examination he showed diplopia, inability to whistle, with marked weakness of the circumoral muscles.

There was no paresis of the vocal cords, but after reading aloud for about twenty minutes his speech became unintelligible. Dysphagia was also present. Motor power in the limbs was practically unimpaired, but fatigue could be easily induced. The deep reflexes were active and the plantar responses flexor in type. There was no myasthenic electrical reaction. The cerebro-spinal fluid was normal. During the next two years there were relapses and remissions in his condition, but in 1926 he was practically well except for an undue fatiguability of his voice. The authors base their diagnosis of epidemic encephalitis in the first case on the initial somnolence and fever. From their description of the cases it would seem possible that in the first case there was an attack of epidemic encephalitis. In the second case there is nothing incompatible with a diagnosis of myasthenia gravis *per se*.

Guillain, Alajouanine and Kalt [12] describe the case of a male, aged 57, who in January, 1925, developed diplopia, which at first was persistent, but later became transitory, appearing especially towards the evening or after fatigue. In October, 1925, variable ptosis, worse always towards the evening, pains in the lumbo-sacral region and fatigue in walking were noticed. Towards the end of a meal mastication became difficult, and there was a certain degree of dysarthria present. He had bilateral ptosis, paresis of the right external rectus and weakness of the masseters. Motor power in the limbs was normal except for a rapid fatiguability. The myasthenic electrical reaction was absent. In addition, the patient showed signs of a slight unilateral Parkinsonism. The second case was that of a male, aged 18, who was taken ill in December, 1922, with diplopia, bilateral ptosis and somnolence, which lasted three weeks. During the following months various myasthenic symptoms developed, such as an abnormal fatiguability of the limbs and masticatory muscles. There is no further note as to the outcome of this case. These authors conclude that in chronic encephalitis muscular weakness may develop, sometimes isolated, sometimes more or less generalized, appearing on or accentuated by muscular effort which closely resembles that seen in myasthenia gravis. Only a careful examination of muscular tonus and the absence of a myasthenic electrical reaction permit a differentiation between the two conditions.

Leenhardt, Reverdiget and Gondard [13] report the case of a boy of 13 in whom Argyll-Robertson pupils, ocular palsies, and double extensor plantar responses and rapid fatiguability of the muscles were noted. This case need not be considered on account of its general lack of resemblance to one of myasthenia gravis.

These few cases can be roughly divided into two groups: (1) Those in which the clinical picture of myasthenia gravis is well represented, but in which the occurrence of a previous attack of epidemic encephalitis is problematical; (2) those in which there is little doubt that the patient had an attack of epidemic encephalitis, but in which the condition that followed lacked several of the true myasthenic phenomena.

The cases described by Wimmer and Vedmand, and Paulian, would seem to fall into the first group, whilst the remainder correspond to the second group. Three points arise from a consideration of these cases:—

(1) Some of the authors lay stress on muscular atrophy as indicative of an underlying encephalitic infection, but in view of the fact that muscular atrophy is by no means rare in true myasthenia gravis, it is obvious that in this connection this sign should not be relied upon.

(2) In the majority of the case reports no mention is made of the presence or absence of paresis of the facial musculature. Wimmer and Vedmand's first case is an exception. It is well recognized that some degree of facial paresis is usual in the course of myasthenia gravis, the orbicularis oculi being most frequently affected, and therefore an intact facial musculature would be evidence tending to contradict the diagnosis of myasthenia gravis.

(3) Most of the authors lay considerable stress on the presence or absence of the myasthenic electrical reaction first described by Jolly. Although its presence is further diagnostic proof, its absence certainly does not militate against this diagnosis.

In reviewing the above cases it is evident that although there are some grounds for occasionally associating myasthenia gravis and epidemic encephalitis, pathological evidence is at present lacking in support of this.

In Grossman's second case, which came to autopsy, the clinical resemblance to myasthenia gravis, was very slight, the bulbar symptoms being terminal.

No case in the literature has been encountered in which there were inflammatory lesions mainly confined to the cord, and a clinical picture resembling myasthenia gravis such as were found in the present case. The nearest approach to such a case is one described by van Bogaert [5].

A male, aged 35, in December, 1925, first experienced dysarthria towards the evening or after much fatigue; this was accompanied by a feeling of constriction in the throat. Four weeks later these symptoms had become aggravated, and movements of the tongue were now difficult. There was also dysphagia, especially with liquids. The patient also noticed a difficulty in holding up his head, and he became subject to attacks of dyspnœa. When examined in January, 1926, he showed definite atrophy and weakness of the circumoral muscles, tongue, masticatory muscles, and those of the neck. There was no fibrillary tremor, except in the tongue. The palatal reflex was present. Speech was bulbar in type and dysphagia was marked. The eyes were normal. No weakness of the limbs or alteration in the reflexes could be detected. There was no sensory loss on the face or limbs. Between January 1 and February 26, the condition gradually progressed, especially with regard to swallowing, speech and mastication. Paresis of the right arm developed, chiefly affecting the muscles of the hand, which showed commencing atrophy. The facial muscles *in toto* were now affected, and the circumoral muscles were so weak that food was retained in the mouth with difficulty. When examined on March 2, gross atrophy and weakness of the tongue and muscles of the face were noted. The speech was unintelligible. The hands presented the picture seen in the Aran-Duchenne type of atrophy. The deep reflexes were absent in the right arm and feebly present in the left. Walking was still possible, and the deep reflexes were conserved with normal plantar flexion. Death took place on April 17, five months from the onset. The main histological findings were: (1) The third nerve nucleus was normal. There was slight perivascular cuffing in the mid-brain. (2) In the pons many of the capillaries were congested. (3 In the dorsal portion of the medulla small neuroglial nodules were present in the neighbourhood of vessels. Perivascular lymphocytosis was present beneath the floor of the fourth ventricle. The twelfth nucleus, Roller's nucleus, the ventral nucleus of the seventh, and dorsal nucleus of the tenth nerve showed chromatolytic changes of varying severity. In the cord, especially in the cervical region, the vessels of the white matter were infiltrated with lymphocytes and plasma cells to a marked degree. The anterior horn cells were diminished in number; some of them showed chromatolytic changes. Weigert-Pal sections were normal. In the dorsal and lumbar regions of the cord the vascular lesions were less intense than at a higher level, and the motor cells were for the most part normal. The author points out that the pathological findings definitely exclude the diagnosis of an ordinary bulbar palsy, with progression of the disease caudally. He contrasts the localization of the inflammatory lesions in the cervical cord with the definite nuclear changes in the medulla. He describes another similar case in which death had not occurred.

Discussion.

(*a*) *Clinical.*—The clinical picture and course of the disease in the present case were considered typical of myasthenia gravis during the life of the patient. The case was seen by several of my colleagues from the point of view of treatment, and this diagnosis was always confirmed. The question of a possible antecedent attack of epidemic encephalitis was considered in the first instance, but no distinctive symptoms suggesting the acute phase of this disease could be elicited. The histological findings in the cord seemed to throw doubt on the validity of the diagnosis, and therefore the possible relationship of the case to epidemic encephalitis came to be considered. As we have seen, the clinical evidence for a myasthenic type of epidemic encephalitis is not very convincing, but at the same time one or two of the published cases are suggestive in this connection. If epidemic encephalitis does actually produce in some cases a picture of myasthenia gravis, one would have expected an increase in the number of cases of this disease. As far as I am aware, there is no published evidence in support of this view. Turning to the pathological side of the question, what interpretation can be placed on the inflammatory changes in the spinal cord? Assuming for the moment that these may be due to epidemic encephalitis, let us consider the clinical symptoms which may result from involvement of the spinal cord in this disease as proved at autopsy. In the first place, in those cases where the changes are widespread, affecting both brain and spinal cord, there may be no symptoms indicating a downward spread of the disease. In other cases, myoclonic movements and root pains are the expression of the cord lesions, and these symptoms may constitute the greater part of the clinical picture. In a much smaller group are found those cases in which histologically the disease has been practically confined to the cord, and in which the clinical picture is that of a myelitis or neuronitis.

It is evident then that the present case considered from the point of view of epidemic encephalitis does not fall into any previously described clinico-pathological group.

(*b*) *Pathological.*—Marked changes in the muscles and viscera were conspicuous by their absence, except in the case of the thymus gland which showed hypertrophy. If the changes in the central nervous system are to be accorded an important rôle in considering the pathogenesis of the present case, then the hypertrophy of the thymus gland must be considered of little account. Buzzard found lymphorrhages in all five cases of myasthenia gravis which he examined. Their absence in the present case suggests the possibility that we are dealing with a type of myasthenia gravis which differs from the classical form. The importance of a careful examination of the muscles and viscera for lymphorrhages was emphasized by Buzzard; this was recognized at an early stage of the investigation of this case and the portions of muscles were in most instances cut in serial section.

When we come to consider the changes in the cord, the first point to be decided is the nature of the perivascular lymphocytosis. Is this comparable with lymphorrhagic invasion of muscles and visceral organs? The presence, however, of other types of cells besides lymphocytes, such as endothelial and plasma cells, accompanied by over reaction of the neuroglia, definitely points to an inflammatory process. Comparison of the sections with those from cases of subacute encephalitis showed that there was little difference between the character of the cells around the vessels. The significance of mucocytic degeneration which was such a marked feature, not only in the cord but in the brain, has already been alluded to. As far as the evidence provided by the literature goes, chronic epidemic encephalitis shows this form of degeneration more markedly than any other disease. In this connection it may be stated that the process was greater in this case of myasthenia gravis than in any of the cases of chronic epidemic encephalitis which were examined. Van Bogaert's case, referred to earlier in this paper, closely resembles the present one, both clinically and pathologically, except with regard to the muscular atrophy and the chromatolytic changes in the nuclei of the seventh, tenth and twelfth cranial nerves, and in the anterior horn cells of the cervical cord. These findings, which seem interdependent, were absent from my case.

Finally, a study of this case from every angle has forced me to the conclusion that there is a form of myasthenia gravis indistinguishable clinically from the classical type of this disease, associated with inflammatory changes in the cord, which may have a causal relationship.

I am indebted to Dr. J. G. Greenfield for several of the references dealing with the subject of mucocytic degeneration.

Summary.

(1) A case of myasthenia gravis is described in which inflammatory changes in the spinal cord with widespread mucocytic degeneration were found.

(2) The possibility of an occasional relationship between myasthenia gravis and epidemic encephalitis is considered.

(3) It is concluded from a study of the case that there is a form of myasthenia gravis associated with inflammatory changes in the central nervous system, the cause of which is at present problematical.

REFERENCES.

[1] Ansalone, G. "Contributo alla istologia patologica della demenza precoce," *Riv. sper. di freniat.*, 1924, **48**, 208.

[2] D'Antona, S. "Sulla genesi della cosi dette zolle di disintegrazione a grappolo," *Rassegna di stud. psichiat.*, 1924, **13**, 501.

[3] Bailey, P., and Shaltenbrand, G. "Die mukŏse Degeneration der Oligodendroglia," *Deutsch. Zeitschr. f. Nervenheilk.*, 1927, **97**, 231.

[4] Barrada, Y. A., and Mott, F. W. "Pathological Findings in the Central Nervous System of a Case of Myasthenia Gravis," *Brain*, 1923, **46**, 237.

[5] Van Bogaert, L. "Étude sur la paralysie labio-glosso-laryngée à évolution subaiguë," *Rev. Neur.*, 1928, **35**, 69.

[6] Buscaino, V. M. "La cause anatomopatologische delle manifestazioni Schizofreniche nella demenza precoce," *Riv. di patol. nerv.*, 1920, **25**, 197.

[7] Buzzard, E. F. "The Clinical History and Post-mortem Examination in Five Cases of Myasthenia Gravis," *Brain*, 1905, **28**, 438.

[8] Ferraro, A. "Acute Swelling of the Oligodendroglia and Grape-like Areas of Disintegration," *Arch. of Neur. and Psych.*, 1928, **20**, 1065.

[9] Greenfield, J. G. "The Pathology of Epidemic Encephalitis," *Journ. of State Medicine*, 1926, **34**, 703.

[10] Grossman, M. "Epidemic Encephalitis simulating Myasthenia gravis," *Journ. of Ment. and Nerv. Dis*,, 1922, **55**, 32.

[11] Grynfeltt, E. "Mucocytes et leur significance," *Compt. Rend. et Mém. de Soc. de Biol.*, 1923, **89**, 1264.

[12] Guillain, G., Alajouanine, Th., et Kalt, M. "La forme myasthénique de l'encéphalite épidémique prolongée," *Rev. neurol.*, 1926, **33**, 39.

[13] Leenhardt, Reverdicet et Gondard. "Un cas de myasthénie postencéphalitique," *Bull. de la Soc. de Sci. Méd. de Montpelier*, 1927, **5**, 226.

[14] Lhermitte, J., et Radovici, A. "Étude sur la dégénération basophile metachromatique dans l'encéphalite épidémique," *Compt. rend. de la Soc. de Biol.*, 1921, **84**, 931.

[15] Lhermitte, J., Kraus, W. M., and Bertillon, F. "Mucin-like Bodies in the Central Nervous System in Epidemic Encephalitis," *Arch. of Neur. and Psych.*, 1924, **12**, 620.

[16] Kescher, M., and Strauss, I. "Myasthenia Gravis," *ibid.*, 1927, **17**, 337.

[17] Mandelbaum, F. S., and Celler, H. L. "A Contribution to the Pathology of Myasthenia Gravis," *Journ. of Exper. Med.*, 1908, **10**, 308.

[18] Pagès, M., Bénoit et Pélissier, G. "La dégénérescence mucocytaire de la Nevroglie," *Languédoc Médical*, 1926, **9**, 1.

[19] Paulian, D. E. "Considérations étiologiques sur la myasthénie et ses relations avec l'encéphalite épidémique chronique," *Rev. Neurol.*, 1926, **33**, 368.

[20] Pélissier, G. "Syndrome Wilsonien consécutif à la névraxite épidémique," *Thèse de Montpelier*, 1924.

[21] Salustri, E. "Sulle cosi dette zolle di disintegrazione a grappolo," *Rev. sper. di freniat.*, 1924, **48**, 255.

[22] Sarbo, A. "Ueber die Encephalitis epidemica, auf Grund der Erfahrungen der Epidemie von 1920," Abst. *Zeit. f. d. ges. Neur. u. Psych.*, 1921, **27**, 509.

[23] Wimmer, A., et Vedmand, H. "Le syndrome myasthénique dans l'encéphalite épidémique chronique," *Rev. Neurol.*, 1926, **33**, 368.